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Project Title:	HZE Radiation: Modulation Of Genetic Effects By RNA Interference Of NHEJ		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	80523-1618	**Congressional District:**	2
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Flight Assignment:	NOTE: Received NCE to 06/01/2009 per PI (7/2008)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bedford, Joel (Colorado State University) Liber, Howard (Colorado State University)		
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Task Description:

For extended missions in space, astronauts will be exposed to a radiation environment unlike any on the earth's surface. Galactic cosmic rays include a particularly worrisome component of energetic iron nuclei which are high atomic number, high energy (HZE) charged particles with linear energy transfer (LET) similar to alpha particles but with very different track structures. While the relative biological effectiveness (RBE) of a particular radiation roughly depends on the LET, the very different track structures for these two radiations may well modulate this RBE even if they have identical average LETs. Further, this "radiation quality" for the exposures actually relevant for astronauts will be modified by interactions with materials in the spacecraft, including their own bodies.
Basic to virtually all relevant biological effects of ionizing radiation is the underlying damage produced in DNA, and the subsequent cellular processing of such damage. The damage can be qualitatively different for different kinds of radiations, and the genetics of the biological systems exposed can greatly affect damage processing and ultimate outcome—the biological effect of concern. The accurate repair of DNA double-strand breaks (DSB) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosome rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. It is important, therefore, to understand cellular responses to DNA damage and to elucidate the mechanisms and phenotypic consequences of DSB misrepair.

Our general aim in this proposal is to examine connections between cellular radiosensitivity phenotypes relevant for carcinogenesis, namely chromosome aberration formation, mutagenesis and telomere fusions, and deficiencies in various components of the non-homologous end joining system, using RNA interference to systematically knockdown gene products known to be involved, either singly or in combination.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

This project utilizes the powerful technique of RNA interference (RNAi) to deplete levels of various components of the non-homologous end joining (NHEJ) double-strand break (DSB) repair system, and to study the effects on HZE radiation-induced chromosomal rearrangements and gene mutation.
The accurate repair of DNA double-strand breaks (DSBs) is critical for the maintenance of genomic integrity and function. Incorrect repair of such lesions results in chromosomal rearrangements and mutations that can lead to cancer and heritable defects in the progeny of irradiated parents. We have focused on the consequent phenotypic effects of faulty repair by examining connections between cellular radiosensitivity phenotypes relevant for carcinogenesis after exposure to IR, and deficiencies in various components of the non-homologous end joining (NHEJ) system. We produced deficiencies of individual components of the DNA-dependent protein kinase (DNA-PK) holoenzyme (Ku86 and the catalytic subunit, DNA-PKcs), both singly and in combination, using RNA interference (RNAi) in human lymphoblastoid cell lines. Exposure of cells exhibiting reduced protein expression to either gamma-rays or 1GeV/nu Fe particles demonstrated differential effects on telomere dysfunction and mutation frequency, as well as differential effects between radiation qualities.

In 2006, we initiated mutagenesis studies with inhibition of Artemis, either alone or in conjunction with DNA-PKcs. The results are quite intriguing. Knockdown of Artemis alone had no effect on spontaneous mutagenesis or on radiation-induced mutagenesis, for either HZE or gamma-rays. However, knockdown of Artemis appeared to prevent the enhancement of mutagenesis that is mediated by inhibition of DNA-PKcs. This suggests that the presence of Artemis is necessary for increased mutagenesis to occur in the absence of DNA-PKcs. We are repeating this experiment; if the result is confirmed, further experiments will be designed to explore the mechanism.

Mutations of the NBS1 gene are responsible for Nijmegen breakage syndrome (NBS), a rare autosomal recessive hereditary disorder that imparts an increased predisposition to development of malignancy. The NBS1 protein is a component of the MRE11/RAD50/NBS1 (MRN) complex that in turn plays a critical role in the regulation of cellular responses to DNA damage and thus the maintenance of chromosomal integrity. NBS has been proposed to function in both NHEJ and HR. Due to the fact that NBS1 is not a component of NHEJ, but still critical for DNA repair, we decided to look at the effects of a deficiency of NBS1 on the consequences of IR. Use of our now standard siRNA technique resulted in a 95% reduction in NBS1 protein 5 days after transfection into WTK1 cells. The cells showed impaired phosphorylation of H2AX, as well as impaired activation of the cell cycle checkpoint regulating kinase, Chk2. These results thus provided evidence that the knockdown was effective on a functional level. We observed an increased level of gamma-ray-induced mutation at two genetic loci. NBS1 deficiency also resulted in a significant increase in telomere association that was independent of radiation exposure. Our results provide the first experimental evidence that NBS1 deficiency in human cells leads to hypermutability and telomere associations, phenotypes that may contribute to the cancer predisposition seen among patients with this disease.

With the exception of Artemis, no individuals null for NHEJ proteins have been identified in the population. This, of course, does not discount the possibility that partial deficiencies may be relevant to issues of human health. Utilizing an RNA interference (RNAi) strategy with siRNA, we have shown that radiation-induced mutation frequencies are affected when DNA-PKcs levels fall below 50% of normal. Furthermore, frequencies of IR-induced telomere-DSB fusion are increased when protein levels of DNA-PKcs are reduced to roughly 50% that of controls. No telomere-telomere fusions were observed under these conditions, but uncapping of telomeres did occur, as evidenced by numerous telomere-DSB fusions, proving these events to be sensitive markers of telomere dysfunction, reflective of underlying repair deficiency. It is noteworthy that no further increase in the frequency of telomere-DSB fusions was seen when the dose was raised from 1.5 to 5 Gy. Such a plateau in the dose response relationship supports the hypothesis that uncapped telomeres are rate-limiting. This study demonstrated that the NHEJ protein DNA-PKcs serves to suppress telomere dysfunction in a manner dependent on protein abundance. Partial deficiencies in DNA-PK relevant to the human condition, if they exist, are more likely to involve haploinsufficiencies, rather than complete loss of function, and would probably involve low penetrance genes that are fixed in the human population at relatively high frequencies. If this notion is correct, the potential impact of inter-individual susceptibility regarding the incidence of telomere-DSB requires examination in the context of human cancer risk.

Our experiments will lead to valuable information regarding CPR question 10.12, “What are the effects of inter-individual diversity?” We hypothesize that NHEJ repair components are major genetic determinants controlling

	individual susceptibility to radiation-induced cancer, as well as heritable effects of exposure. Therefore, naturally occurring deficiencies of these proteins within the population will put these individuals at increased risk. We also anticipate addressing CPR question 10.07, “What are the cancer risks in humans from spaceflight?” We anticipate that our characterization of telomere dysfunction, chromosome aberration formation and gene mutation will serve as surrogate endpoints for predicting cancer risks.
Bibliography Type:	Description: (Last Updated: 12/10/2025)
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